

三角函数公式表

同角三角函数的基本关系式

倒数关系：

$$\begin{aligned}\tan \alpha &\cdot \cot \alpha = 1 \\ \sin \alpha &\cdot \csc \alpha = 1 \\ \cos \alpha &\cdot \sec \alpha = 1\end{aligned}$$

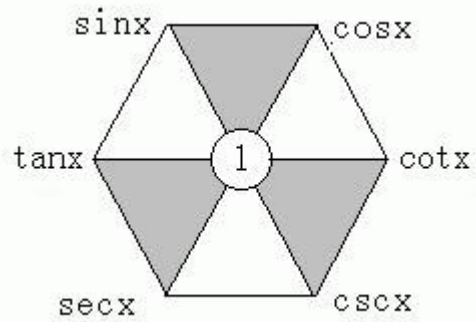
商的关系：

$$\begin{aligned}\sin \alpha / \cos \alpha &= \tan \alpha = \sec \alpha / \csc \alpha \\ \cos \alpha / \sin \alpha &= \cot \alpha = \csc \alpha / \sec \alpha\end{aligned}$$

平方关系：

$$\begin{aligned}\sin^2 \alpha + \cos^2 \alpha &= 1 \\ 1 + \tan^2 \alpha &= \sec^2 \alpha \\ 1 + \cot^2 \alpha &= \csc^2 \alpha\end{aligned}$$

（六边形记忆法：图形结构“上弦中切下割，左正右余中间1”；记忆方法“对角线上两个函数的积为1；阴影三角形上两顶点的三角函数值的平方和等于下顶点的三角函数值的平方；任意一顶点的三角函数值等于相邻两个顶点的三角函数值的乘积。”）



诱导公式（口诀：奇变偶不变，符号看象限。）

$\sin(-\alpha) = -\sin \alpha$	$\cos(-\alpha) = \cos \alpha$	$\tan(-\alpha) = -\tan \alpha$	$\cot(-\alpha) = -\cot \alpha$
$\sin(\pi/2 - \alpha) = \cos \alpha$	$\sin(\pi - \alpha) = \sin \alpha$	$\sin(3\pi/2 - \alpha) = -\cos \alpha$	$\sin(2\pi - \alpha) = -\sin \alpha$
$\cos(\pi/2 - \alpha) = \sin \alpha$	$\cos(\pi - \alpha) = -\cos \alpha$	$\cos(3\pi/2 - \alpha) = -\sin \alpha$	$\cos(2\pi - \alpha) = \cos \alpha$
$\tan(\pi/2 - \alpha) = \cot \alpha$	$\tan(\pi - \alpha) = -\tan \alpha$	$\tan(3\pi/2 - \alpha) = \cot \alpha$	$\tan(2\pi - \alpha) = -\tan \alpha$
$\cot(\pi/2 - \alpha) = \tan \alpha$	$\cot(\pi - \alpha) = -\cot \alpha$	$\cot(3\pi/2 - \alpha) = \tan \alpha$	$\cot(2\pi - \alpha) = -\cot \alpha$
$\sin(\pi/2 + \alpha) = \cos \alpha$	$\sin(\pi + \alpha) = -\sin \alpha$	$\sin(3\pi/2 + \alpha) = -\cos \alpha$	$\sin(2k\pi + \alpha) = \sin \alpha$
$\cos(\pi/2 + \alpha) = -\sin \alpha$	$\cos(\pi + \alpha) = -\cos \alpha$	$\cos(3\pi/2 + \alpha) = \sin \alpha$	$\cos(2k\pi + \alpha) = \cos \alpha$
$\tan(\pi/2 + \alpha) = -\cot \alpha$	$\tan(\pi + \alpha) = \tan \alpha$	$\tan(3\pi/2 + \alpha) = -\cot \alpha$	$\tan(2k\pi + \alpha) = \tan \alpha$
$\cot(\pi/2 + \alpha) = -\tan \alpha$	$\cot(\pi + \alpha) = \cot \alpha$	$\cot(3\pi/2 + \alpha) = -\tan \alpha$	$\cot(2k\pi + \alpha) = \cot \alpha$

(其中 $k \in \mathbb{Z}$)

两角和与差的三角函数公式

$$\begin{aligned}\sin(\alpha + \beta) &= \sin \alpha \cos \beta + \cos \alpha \sin \beta \\ \sin(\alpha - \beta) &= \sin \alpha \cos \beta - \cos \alpha \sin \beta \\ \cos(\alpha + \beta) &= \cos \alpha \cos \beta - \sin \alpha \sin \beta \\ \cos(\alpha - \beta) &= \cos \alpha \cos \beta + \sin \alpha \sin \beta\end{aligned}$$

$$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \cdot \tan \beta}$$

$$\tan(\alpha - \beta) = \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \cdot \tan \beta}$$

万能公式

$$\sin \alpha = \frac{2 \tan(\alpha/2)}{1 + \tan^2(\alpha/2)}$$

$$\cos \alpha = \frac{1 - \tan^2(\alpha/2)}{1 + \tan^2(\alpha/2)}$$

$$\tan \alpha = \frac{2 \tan(\alpha/2)}{1 - \tan^2(\alpha/2)}$$

半角的正弦、余弦和正切公式

$$\sin \frac{\alpha}{2} = \pm \sqrt{\frac{1 - \cos \alpha}{2}}$$

$$\cos \frac{\alpha}{2} = \pm \sqrt{\frac{1 + \cos \alpha}{2}}$$

$$\tan \frac{\alpha}{2} = \pm \sqrt{\frac{1 - \cos \alpha}{1 + \cos \alpha}} = \frac{1 - \cos \alpha}{\sin \alpha} = \frac{\sin \alpha}{1 + \cos \alpha}$$

三角函数的降幂公式

$$\sin^2 \alpha = \frac{1 - \cos 2\alpha}{2}$$

$$\cos^2 \alpha = \frac{1 + \cos 2\alpha}{2}$$

二倍角的正弦、余弦和正切公式

$$\sin 2\alpha = 2\sin \alpha \cos \alpha$$

$$\cos 2\alpha = \cos^2 \alpha - \sin^2 \alpha = 2\cos^2 \alpha - 1 = 1 - 2\sin^2 \alpha$$

$$\tan 2\alpha = \frac{2\tan \alpha}{1 - \tan^2 \alpha}$$

三倍角的正弦、余弦和正切公式

$$\sin 3\alpha = 3\sin \alpha - 4\sin^3 \alpha$$

$$\cos 3\alpha = 4\cos^3 \alpha - 3\cos \alpha$$

$$\tan 3\alpha = \frac{3\tan \alpha - \tan^3 \alpha}{1 - 3\tan^2 \alpha}$$

三角函数的和差化积公式

$$\sin \alpha + \sin \beta = 2\sin \frac{\alpha + \beta}{2} \cdot \cos \frac{\alpha - \beta}{2}$$

$$\sin \alpha - \sin \beta = 2\cos \frac{\alpha + \beta}{2} \cdot \sin \frac{\alpha - \beta}{2}$$

$$\cos \alpha + \cos \beta = 2\cos \frac{\alpha + \beta}{2} \cdot \cos \frac{\alpha - \beta}{2}$$

$$\cos \alpha - \cos \beta = -2\sin \frac{\alpha + \beta}{2} \cdot \sin \frac{\alpha - \beta}{2}$$

三角函数的积化和差公式

$$\sin \alpha \cdot \cos \beta = \frac{1}{2} [\sin (\alpha + \beta) + \sin (\alpha - \beta)]$$

$$\cos \alpha \cdot \sin \beta = \frac{1}{2} [\sin (\alpha + \beta) - \sin (\alpha - \beta)]$$

$$\cos \alpha \cdot \cos \beta = \frac{1}{2} [\cos (\alpha + \beta) + \cos (\alpha - \beta)]$$

$$\sin \alpha \cdot \sin \beta = \frac{1}{2} [\cos (\alpha + \beta) - \cos (\alpha - \beta)]$$

化 $a\sin \alpha \pm b\cos \alpha$ 为一个角的一个三角函数的形式（辅助角的三角函数的公式）

$$a\sin x \pm b\cos x = \sqrt{a^2 + b^2} \sin(x \pm \phi)$$

（其中 ϕ 角所在象限由 a 、 b 的符号确定， ϕ 角的值由 $\tan \phi = \frac{b}{a}$ 确定。）

特殊角的三角函数值表

三角函数 \ 角	0°	30°	45°	60°	90°
$\sin \alpha$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
$\cos \alpha$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
$\tan \alpha$	0	$\frac{\sqrt{3}}{3}$	1	$\sqrt{3}$	不存在
$\cot \alpha$	不存在	$\sqrt{3}$	1	$\frac{\sqrt{3}}{3}$	0